Building Materials

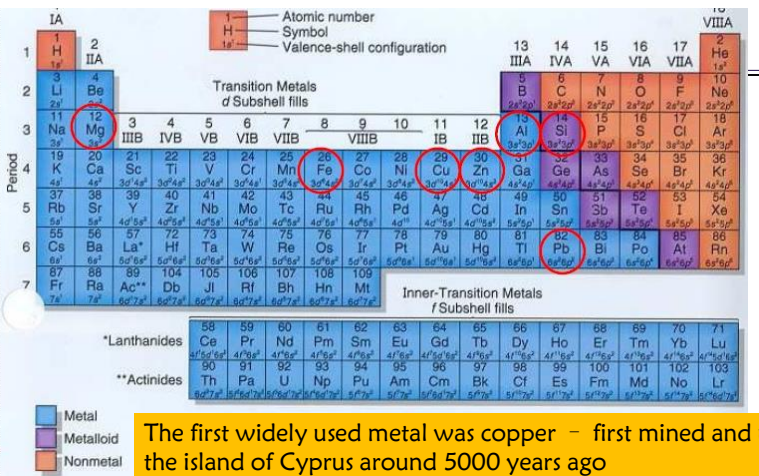
Wei, Ya （魏亚）
Department of Civil Engineering

Tel.： +86-10-62771646
Email：yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood
Chapter Eight Composites

Chapter Three: Metal

1. **Introduction**
2. **Crystalline structure**
3. **Production and Alloys**
4. **Mechanical properties**
5. **Strengthening mechanisms**
6. **Corrosion**
7. **Engineering application**



The first widely used metal was copper – first mined and used on the island of Cyprus around 5000 years ago

- Low melting point
- Easily extracted from its ores

Aluminium (most common metal in the earth's crust)

- Difficult and expensive to extract
- Now widely used due to technological advances

3.1 Introduction



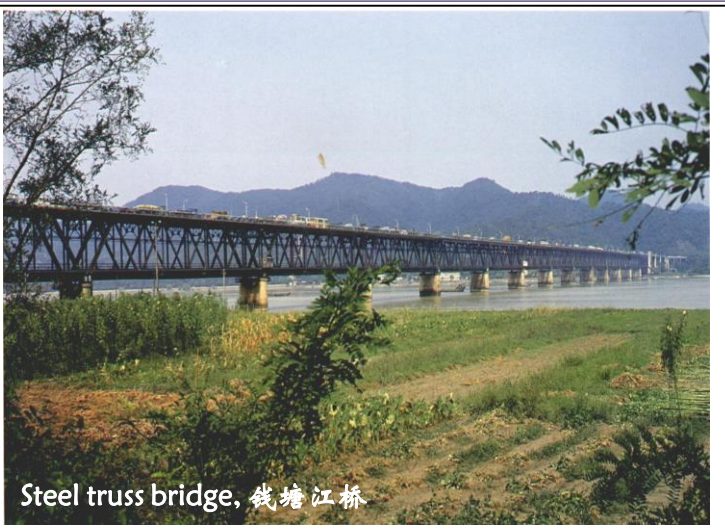
Reinforcement, rebar, mould, bracing member

3.1 Introduction



Suspension cables, Akashi bridge (Japan)

3.1 Introduction



Steel truss bridge, 钱塘江桥

3.1 Introduction

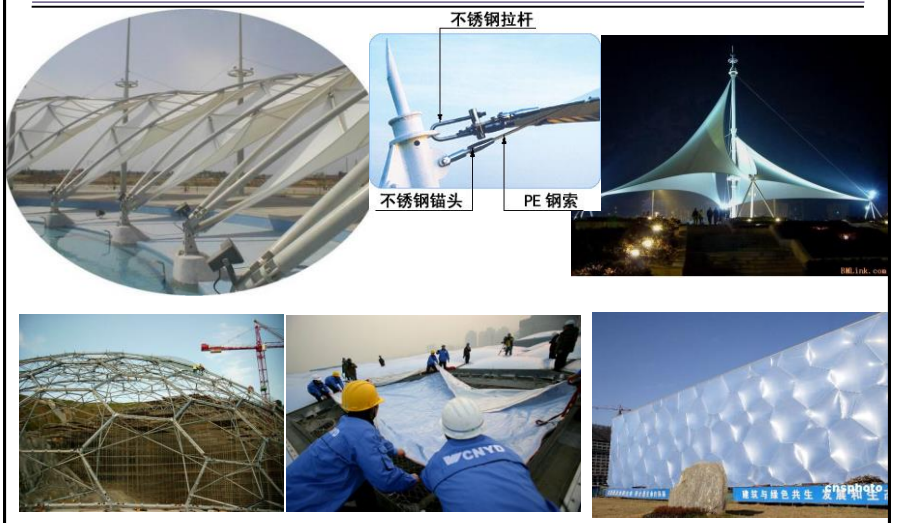
World Trade Center, New York, US



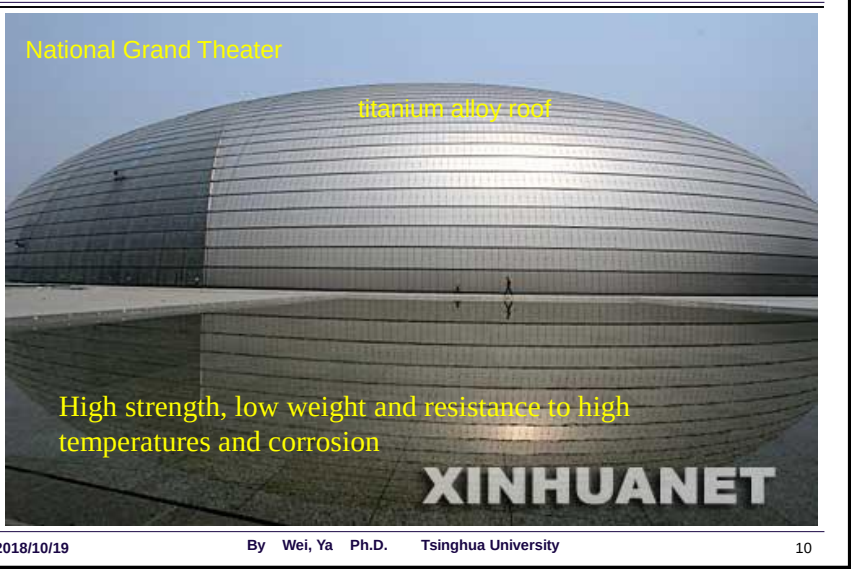
广州电视塔钢结构



3.1 Introduction



3.1 Introduction



3.1 Introduction

Metal	Global demand (million tons)	Price ($ / ton)	Market (billion $)
Steel	730	400	292
Aluminium	20	1,500	30
Stainless steel	13	2,000	26
Titanium	0.056	35,000	1.96
Nickel	1	12,000	12
Copper	10	3,055	30

3.1 Introduction

Structural

- Reinforced concrete
- Prestressed concrete
- Steel structure

functional

- Roof materials
- Decoration materials

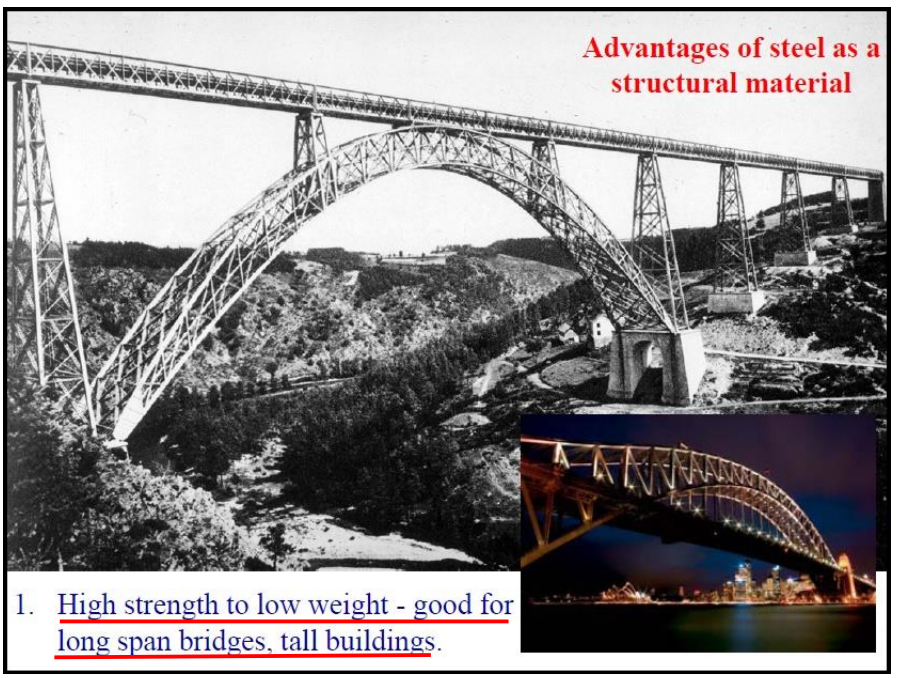
3.1 Introduction

Advantage:

1. High tensile strength (low carbon steel 180~200MPa； concrete2~6MPa)
3. Very strong and flexible
2. Structures can be prefabricated (welded) in large sections, and lifted into place with cranes and bolted together- buildings can be erected quickly.

Disadvantage:

1. poor fire resistance
2. low structure stiffness
3. susceptible to corrosion， high cost for maintenance
4. Energy intensive to produce





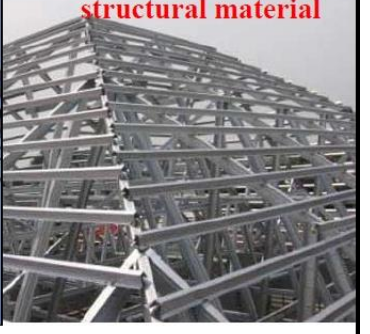
Advantages of steel as a structural material



1. Lightweight compared to concrete - can be handled and transported, and prefabricated.
2. Properly maintained have a long life.
3. Uniformity properties do not change with time.
4. A ductile material, does not fail suddenly, but gives visible evidence of failure by large deflections.



Advantages of steel as a structural material



1. Additions and alterations can be made easily.
2. They can be erected at a faster rate compared to reinforced concrete.
3. Steel has the highest scrap value.
4. Can be even reuse on demolition.





1. When placed in exposed conditions, are subjected to corrosion. They require painting, hence induce high maintenance cost.



2. Needs fire proof treatment, which increase cost.

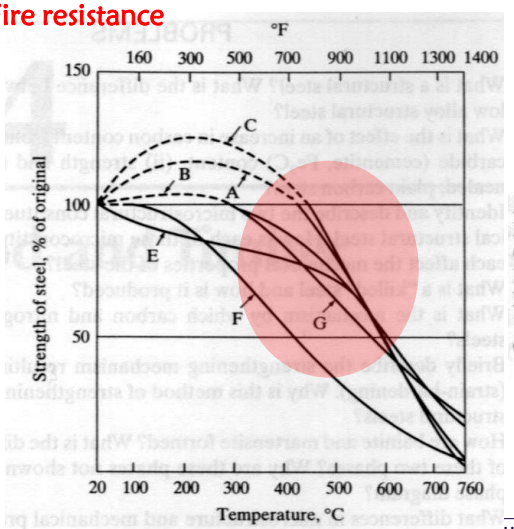
3.1 Introduction



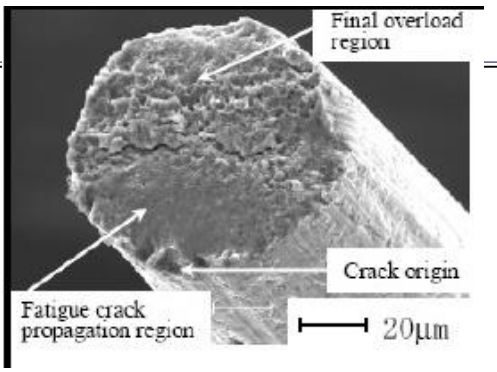


3.1 Introduction

Fire resistance



A Hot-rolled high-yield bars (yield)
B Cold-worked bar (yield)
C Mild-steel bar (yield)
E Hot-rolled steel (yield strength)
F Cold-drawn prestressing steel (tensile strength)
G High-strength alloy-steel bars (Tensile strength)



3. Fatigue – strength reduced if large number of stress reversals.

3.1 Introduction





Pier in the sea



原西直门引桥护栏，北京

Chapter Three: Metal

1. Introduction
2. Crystalline structure
3. Production and Alloys
4. Mechanical properties
5. Strengthening mechanisms
6. Corrosion
7. Engineering application

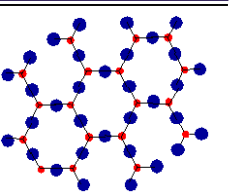
3.2 Crystalline structure

- Metals occur as crystal lattices

Crystalline materials...
- atoms pack in periodic
- typical of: -metals
  -many ceramics
  -some polymers

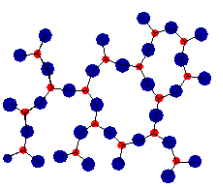
Noncrystalline materials...
- atoms have no periodic packing
- occurs for: -complex structures
  -rapid cooling

"Amorphous" = Noncrystalline



crystalline SiO_2

Si Oxygen



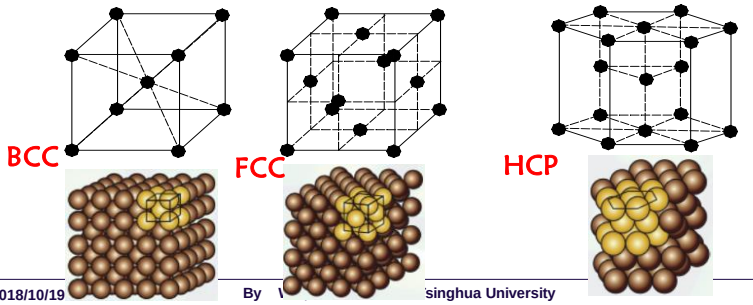
noncrystalline SiO_2

3.2 Crystalline structure

Lattice Structure

The cations in metals can arrange themselves in a number of different periodic arrangements. The most common lattice arrangements are the following:

- Body-centered cubic packing of the atoms (BCC) 体心结构
- Face-centered cubic packing of the atoms (FCC) 面心结构
- Hexagonally close packing of the atoms (HCP) 密排六方结构

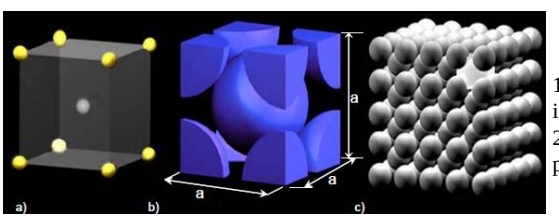


3.2 Crystalline structure

Lattice Structure

1.Body-Centered Cubic

- The body-centered cubic unit cell has atoms at each of the eight corners of a cube plus one atom in the center of the cube.
- Since 8 of its atoms are shared, it is said to have a coordination number of 8.

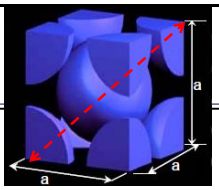


BCC unit cell: a) whole atoms; b) with sectioned atoms at unit cell vertices; and c) periodic repetition of the unit cell.



1. How many complete atoms in this structure?
2. How close are these atoms packed?

3.2 Crystalline structure



Lattice Structure

1.Body-Centered Cubic

In the BCC unit cell, the diagonal of the unit cell can be seen to be equal to 4r, where r is that atomic radius of each metal cation. The side length of the unit cell is denoted by the lattice parameter "a". The diagonal of the unit cell can be expressed as follows:

diagonal: $4r = \left(a^2 + a^2 + a^2\right)^{1/2} \Rightarrow a = \frac{4\sqrt{3}r}{3} = 2.3094\, r$

The volume of the unit cell Vcell is: $V_{cell} = a^3 = 12.317\, r^3$

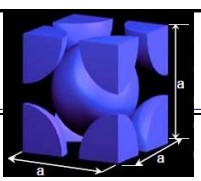
In turn, the volume of the two complete atoms within the BCC unit cell is:

$$V_{atoms} = 2\left(\frac{4}{3}\right)\pi r^3 = 8.3775\, r^3$$

The so-called atomic packing factor (APF) for the unit cell is simply the ratio of the atomic volume to the unit cell volume.

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{8.3775\, r^3}{12.317\, r^3} = 0.68$$

3.2 Crystalline structure



Lattice Structure

1.Body-Centered Cubic

If one knows the atomic radius of metal cations in a lattice structure, the mass density of a given metal can be estimated from the unit cell as follows:

$$\rho = \frac{n_{atoms} * atomic - mass}{V_{cell}}$$

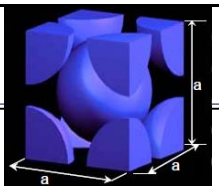
Example: Estimate the mass density of BCC iron. The atomic number of iron is 26, and the atomic radius of iron cations is 0.1241nm.

Solution: The mass of a single iron atom is roughly equal to that of 26 protons and 26 neutrons.

$$\rho = \frac{2 * 26 * \left(1.673 \times 10^{-27} kg + 1.675 \times 10^{-27} kg\right)}{12.32\left(0.1241 \times 10^{-9}\right)^3} = 7408\ kg/m^3$$

the actual mass density of iron 7900 kg/m3.

3.2 Crystalline structure

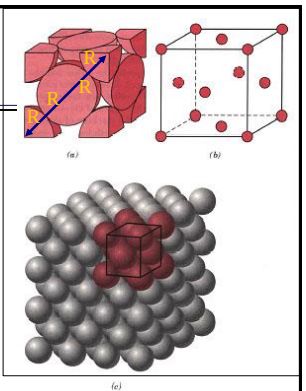


Lattice Structure

1.Body-Centered Cubic

- Some of the materials that have a bcc structure include lithium, sodium, potassium, chromium and tungsten.
- Metals which have a bcc structure are usually harder and less malleable than close-packed metals such as gold.
- When the metal is deformed, the planes of atoms must slip over each other, and this is more difficult in the bcc structure.

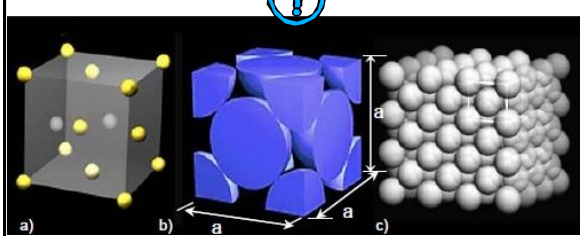
3.2 Crystalline structure



Lattice Structure

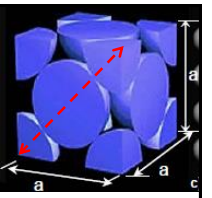
2.Face-Centered Cubic

- The face centered cubic structure has atoms located at each of the corners and the centers of all the cubic faces.



FCC unit cell: a) whole atoms; b) with sectioned atoms at unit cell vertices; and c) periodic repetition of the unit cell.

3.2 Crystalline structure



Lattice Structure

2.Face-Centered Cubic

Within any of the six faces of the FCC unit cell one can observe that the diagonal length is as follows:

Diagonal within face of cell: $4r = \sqrt{2}a \Rightarrow a = 2\sqrt{2}r$

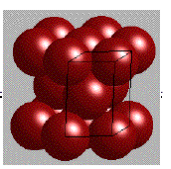
The volume of the FCC unit cell is: $V_{cell} = a^3 = \left(2\sqrt{2}r\right)^3 = 16\sqrt{2}r^3$

For the FCC, since it contains 4 atoms, the APF can be computed as follows:

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{4 * \frac{4}{3}\pi r^3}{16\sqrt{2}r^3} = 0.74$$

- The atomic packing factor in the FCC lattice is greater than that in the BCC.
- Some of the common metals that have the fcc structure include aluminum, copper, gold, lead, nickel, and silver.

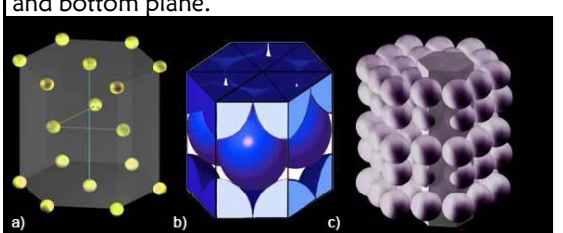
3.2 Crystalline structure



Lattice Structure

3.Hexagonally Close Packing (HCP)

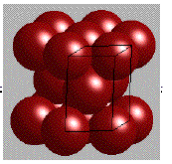
- The hcp structure has three layers of atoms. The top and bottom layer, there are six atoms that arrange themselves in the shape of a hexagon and a seventh atom that sits in the middle of the hexagon.
- The middle layer has three atoms nestle in the triangular "grooves" of the top and bottom plane.



a) unit cell with full atoms at vertices; b) with sectioned atoms at vertices; and c) periodic repetition of the unit cells.

How many complete atoms in the HCP unit cell?

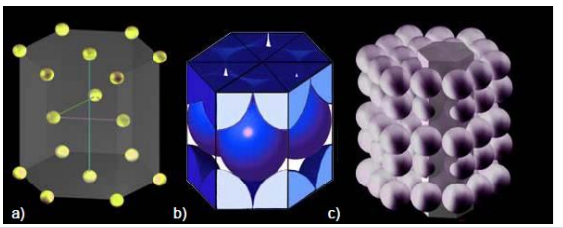
3.2 Crystalline structure



Lattice Structure

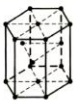
3.Hexagonally Close Packing (HCP)

- There are six atoms in the hcp unit cell. Each of the 12 atoms in the corners of the top and bottom layers contribute 1/6 atom to the unit cell, the two atoms in the center of the hexagon of both the top and bottom layers each contribute atom and each of the three atom in the middle layer contribute 1 atom.
- The packing factor is 0.74, which is the same as the fcc unit cell. The hcp structure is very common for elemental metals and some examples include magnesium, titanium, and zinc.



3.2 Crystalline structure

Lattice Structure

	Lattice structure	Unit cell schematic	Ping-pong ball model	Number of nearest neighbors	Packing efficiency	Typical metals
a	Simple cubic			6	52%	None
b	Body-centered cubic			8	68%	Fe, Cr, Mn, Cb, W, Ta, Ti, V, Na, K
c	Face-centered cubic			12	74%	Fe, Al, Cu, Ni, Ca, Au, Ag, Pb, Pt
d	Hexagonal close-packed			12	74%	Be, Cd, Mg, Zn, Zr

3.2 Crystalline structure

Lattice Structure

Metal	Crystal Structure	Atomic Radius (nm)
Aluminum	FCC	0.1431
Chromium	BCC	0.1249
Cobalt	HCP	0.1253
Copper	FCC	0.1278
Gold	FCC	0.1442
Lead	FCC	0.1750

3.2 Crystalline structure

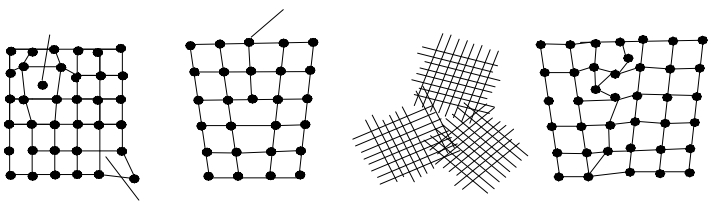
Imperfections

Theoretical strength would be much larger than the actual strengths that metals display.

Defects in the lattice are the explanation for this discrepancy.

These defects are generally classified as follows:

1. zero-dimensional point defects;
2. one-dimensional line defects (dislocations); and
3. two-dimensional surface defects.



1. Introduction
2. Crystalline structure
3. Production and Alloys
4. Mechanical properties
5. Strengthening mechanisms
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7. Engineering application

3.3 Production and Alloys

Raw materials

Minerals occur naturally in Earth's crust



Manganese ore



Gold ore

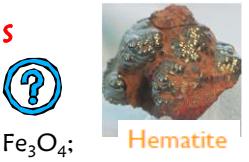


Lead ore

3.3 Production and Alloys

Simplified scheme of a blast furnace

Raw materials


Hematite

Iron ores:

oxides: magnetite Fe_3O_4;
hematite Fe_2O_3;

limonite: $FeO(OH).x\ H_2O$

sulphide: pyrite FeS_2

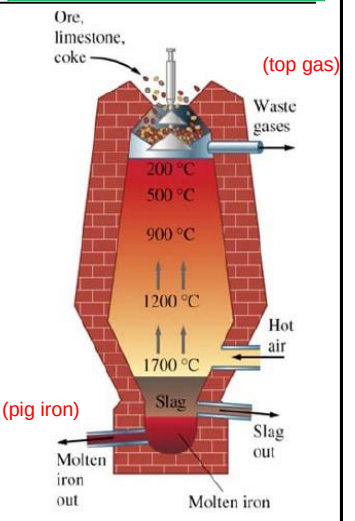
carbonate: siderite $FeCO_3$


Magnetite

Coke, limestone


Pyrite (Fool's Gold)

20~40m



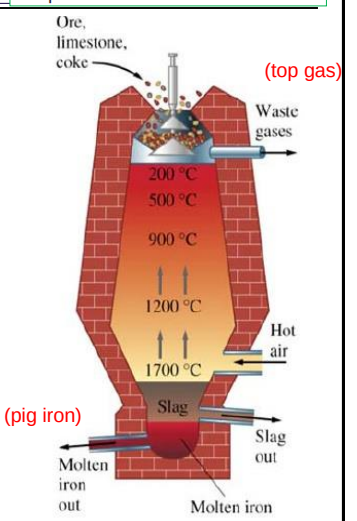
3.3 Production and Alloys

Simplified scheme of a blast furnace

Raw materials

To produce 1 ton of pig iron, it takes about 1.3 tons of iron ore, 0.6 ton kg of coke, 0.4 tons of limestone, 7.3 kg of air, 22 tons of water, and $27x10^6$ BTU of heat.

20~40m



3.3 Production and Alloys

Inside the blast furnace…

- The coke (from coal) reacts with oxygen to produce carbon monoxide (CO):

$$2\,C + O_2 \rightarrow 2\,CO$$

- Carbon monoxide reacts with iron ore to become carbon dioxide (CO_2)

$$3\,CO + Fe_2O_3 \rightarrow 2\,Fe + 3\,CO_2$$

- Limestone ($CaCO_3$) is used as a solvent to help removing the impurities, such as silicon dioxide in the ore

$$CaCO_3 \rightarrow CaO + CO_2$$
$$CaO + SiO_2 \rightarrow CaSiO_3$$ ← slag



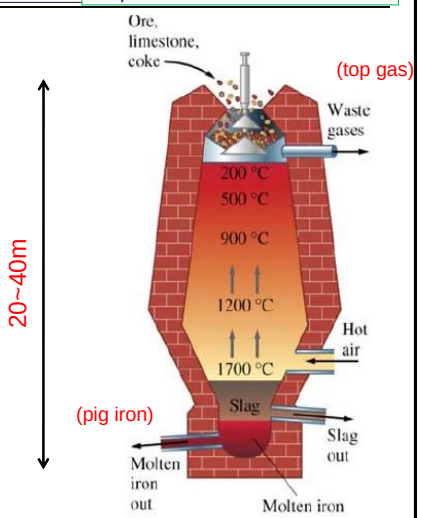
- The molten slag is lighter than the molten iron, so it floats on the top and can be drawn off through an opening at the bottom of the furnace – this can later be used as cement replacement material in concrete (Recall GGBFS = Ground Granulated Blast Furnace Slag)

3.3 Production and Alloys

Raw materials

Pig iron obtained from the blast furnace cannot be used by its own, due to its high carbon content (about 3.5-4%). It has to be processed further to reduce the amount of carbon and to remove other impurities.

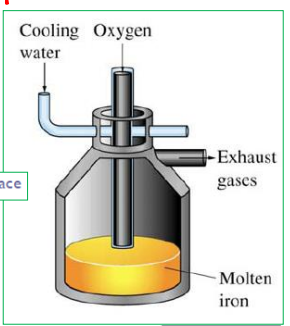
Simplified scheme of a blast furnace



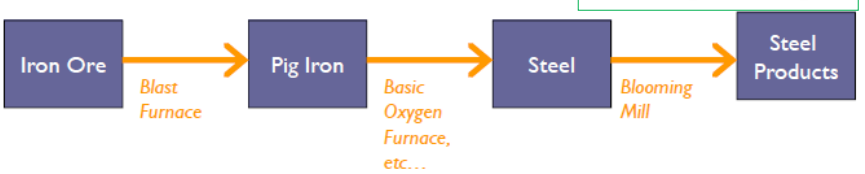
3.3 Production and Alloys

The manufacturing of steel consists of 3 main phases

- Reducing Iron Ore to Pig Iron
- Refining Pig Iron to Steel
- Forming Steel into products



Basic Oxygen Furnace

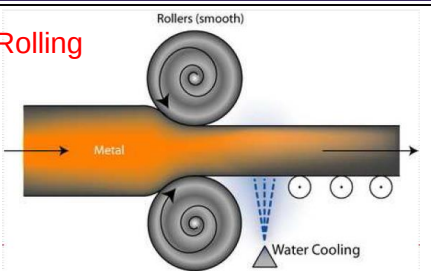


3.3 Production and Alloys

casting



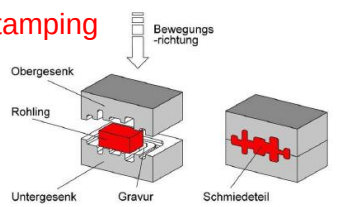
Rolling



Forging

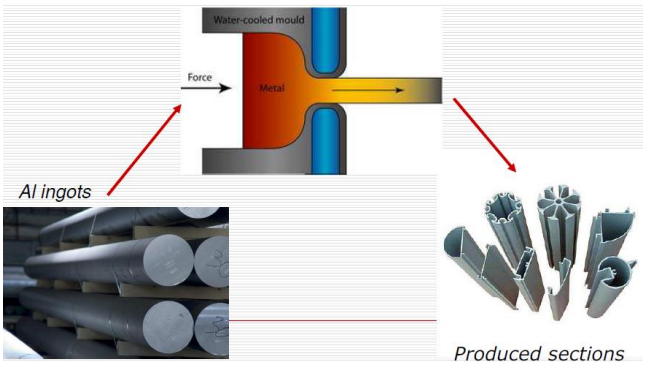


Stamping



3.3 Production and Alloys

Extrusion



3.3 Production and Alloys

Iron: Iron is an element and can be pure.

Cast iron: Iron that contains about as much carbon as it can hold which is about 4%.

Wrought iron: Iron that contains carbon less than 0.02%.

Steel: Iron with a bit of carbon in it— generally less than 1.5%.

Pig iron: Raw iron, the immediate product of melting iron ore with coke and limestone in a blast furnace. Pig iron has a very high carbon content, typically 4-5%, which makes it very brittle and not very useful directly as a material.

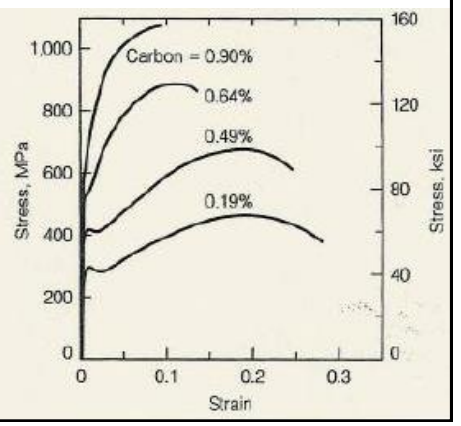
3.3 Production and Alloys

Role of Carbon

carbon atoms replace or enter into solution among the lattice structure of the iron atoms and limit the slip planes in the lattice

Results in changes in:

- Carbon influences:
  - ↑ strength, ↑ hardness
  - ↓ ductility, ↓ toughness



3.3 Production and Alloys

Types of steel

Steel

- Low carbon steel
- Medium carbon steel
- High carbon steel (tool steels)



Alloy Steels: higher content of other elements for: higher corrosion resistance, hardness, strength….

- Stainless steel

3.3 Production and Alloys

Steel

Low Carbon Steel: Contain 0.05% -0.32% carbon

1. Tough, ductile and malleable
2. Easily joined and welded
3. Poor resistance to corrosion
4. Often used a general purpose material



- Nails, screws, car bodies,
- Structural Steel used in the construction industry





3.3 Production and Alloys

Steel

Medium Carbon Steel: Contains 0.35% - 0.5% of carbon Offer more strength and hardness BUT less ductile and malleable Structural steel, rails and garden tools.







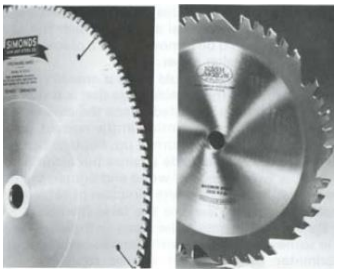
3.3 Production and Alloys

Steel

High Carbon Steel: Also known as ‘tool steel’, contain 0.55%-1.5% carbon

Very hard but offers Higher Strength Less ductile and less malleable

Hand tools (chisels)
Saw blades





3.3 Production and Alloys

Cast iron:

- Cast iron is produced by reheating pig iron and remove some of the impurities. It contains about 2-4% of carbon
- It can be cast into molds
- It is brittle and best used in compression rather than tension
- Common applications are pipes and fittings.
- Cast iron is difficult to weld.
- It is cheap, hard, brittle, poor tensile strength



3.3 Production and Alloys

Pure Metals and Alloys:

Metal that are not mixed with any other materials are known as pure metals. Metals listed in the Periodic Table are pure Metals. E.g. Iron (Fe), Copper (Cu) and Zinc (Zn).

Alloys are **mixtures of two or more metals formed together** with other elements/materials to create new metals with improved properties and characteristics.

E.g. Brass (Copper and Zinc),
Stainless steel (steel and chromium)

Alloy = metal A + metal B + ⋯ + other elements

3.3 Production and Alloys



Alloy steel

Alloy steel: A generic term for steels which are alloyed with elements other than carbon. Why alloys?

- Alloy steel is steel alloyed with a variety of elements in amounts of between 1 and 50% by weight to improve its mechanical properties.
- Alloy steels have greater strength, hardness, hot hardness, wear resistance, or toughness compared to carbon steel.
- However, they may require heat treatment to achieve such properties.



3.3 Production and Alloys

Alloy steel

Stainless Steel

- a steel alloy with a minimum of 10.5 or 11% chromium content by mass.
- Stainless steel does not stain, corrode, or rust as easily as ordinary steel (it *stains less, but it is not stain-proof).*
- Stainless steel is used where the properties of steel, and resistance to corrosion are required.
- Stainless steel differs from carbon steel by the amount of chromium present.
- Carbon steel rusts when exposed to air and moisture. Stainless steels contain sufficient chromium to form a passive film of chromium oxide, which prevents further surface corrosion and blocks corrosion from spreading into the metal's internal structure.

3.3 Production and Alloys

Alloy steel

- Copper and chromium are added to produce weather resistant steels that do not require corrosion protection.
- Main design property of structural steel is based on the yield strength of the steel, but other properties including ductility, toughness, impact resistance and weldability are also important.





3.3 Production and Alloys

Aluminium alloys

Low density, high corrosion resistance (passivation by Al_2O_3), good electric and heat conductor
Rather high price (compared to steel)

Pure aluminium has low strength – Al is used only in alloys

Al – Mn good formability, Mn to 1.5 % to make drink cans



Chapter Three

1. Introduction
2. Crystalline structure
3. Production and Alloys
4. Mechanical properties
5. Strengthening mechanisms
6. Corrosion
7. Engineering application

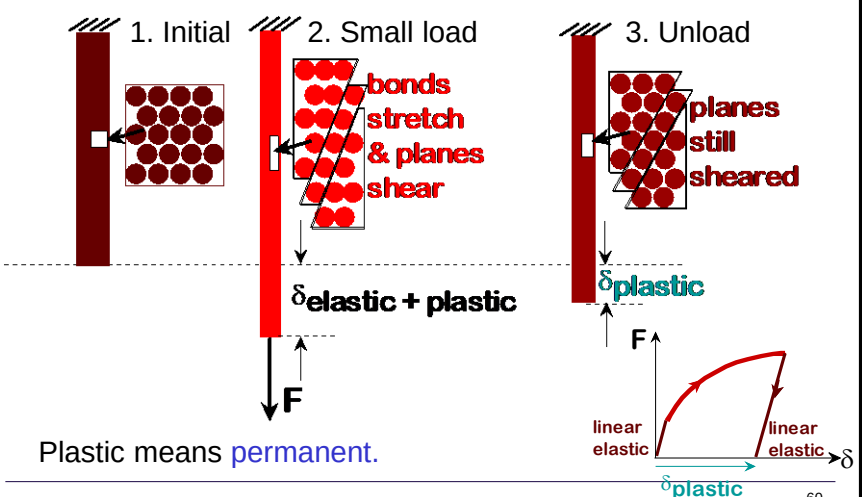
3.4 Mechanical Properties

- Stiffness - Elastic Modulus or Young's Modulus (MPa)
- Strength - Yield, Ultimate, Fracture, Measured as stress (MPa)
- Hardness - Resistance to abrasion (Various scales, e.g.; Rockwell, Brinell, Vickers.)
- Ductility - Measure of ability to deform plastically without fracture - Elongation, Area Reduction, Fracture Strain - (no units or mm/mm)
- Toughness - Resilience - Measure of ability to absorb energy (J/m^3).

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3.4 Mechanical Properties

Tensile Strength

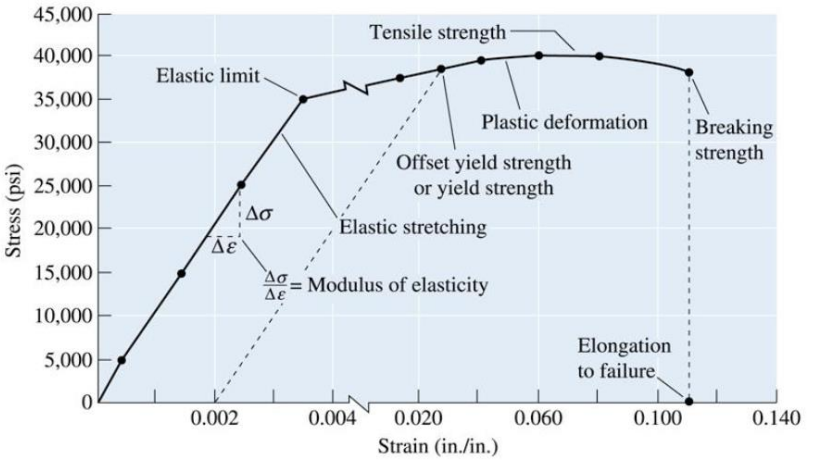


Plastic means permanent.

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3.4 Mechanical Properties

Stress Strain Curve for Aluminum

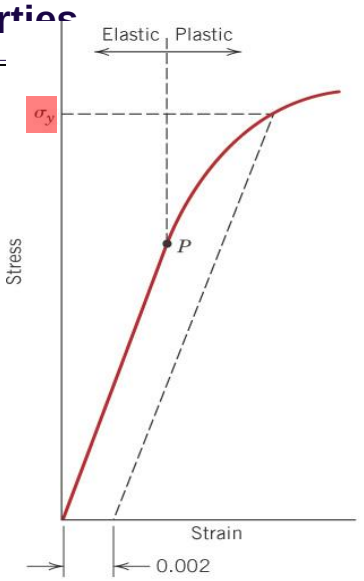


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3.4 Mechanical Properties

Tensile Strength

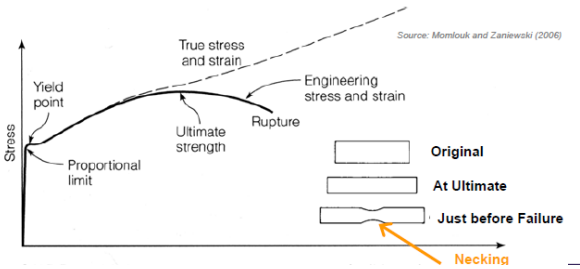
Typical stress-strain behavior for a metal showing elastic and plastic deformations, the proportional limit P and the yield strength σ_y, as determined using the 0.002 strain offset method (where there is noticeable plastic deformation). P is the gradual elastic to plastic transition.



3.4 Mechanical Properties

Necking

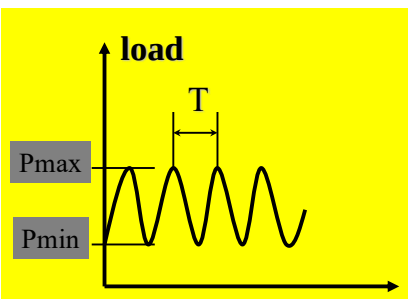
- After the ultimate strength, the tensile test member exhibit "necking" behavior in which the area of the test piece decreases as the deformation increases with the corresponding decrease in load.
- Engineering Stress = F/A $_{original}$, we get a decrease in stress from ultimate to rupture.
- True Stress = F/A $_{actual}$, we get an increase in stress (dashed line) instead.

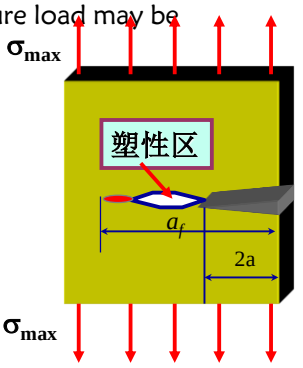


3.4 Mechanical Properties

Fatigue

- Fatigue is damage caused by repeated fluctuations of stress leading to gradual cracking of a structural element.
- Failure occurs through initiation and propagation of a crack that starts at a fault or structural discontinuity and the failure load may be well below its static value.

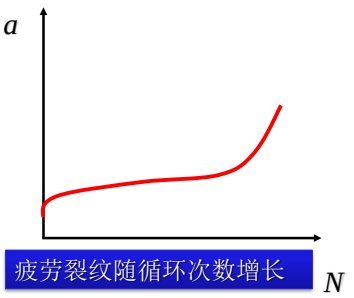


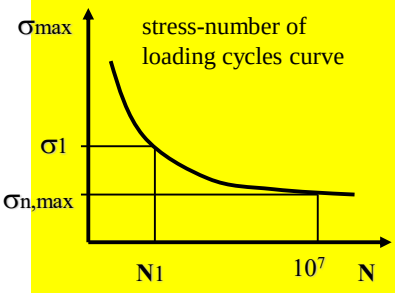


3.4 Mechanical Properties

Fatigue

- Welded connections have the greatest effect on the fatigue strength of steel structure, bolted connections do not reduce the strength under fatigue loading.
- To help avoid fatigue failure, stress concentrations and abrupt changes of section are avoided in regions of tensile stress.





Summary

1. Crystalline Structure

BCC, FCC, HCP

Atomic Packing Factor

2. Production and Alloy

Production Process

Carbon Content, Why Alloy

3. Mechanical Properties

Yield Strength

Fatigue

QUESTIONS?

